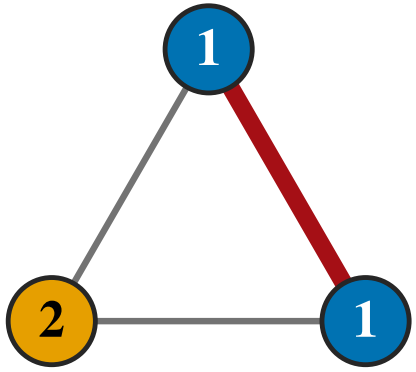
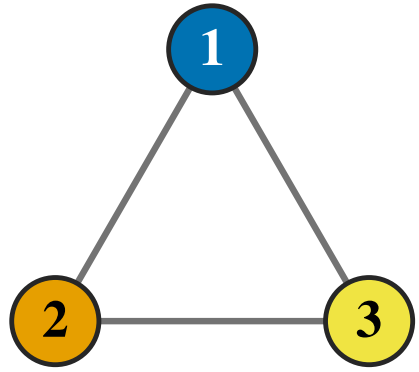
(a)

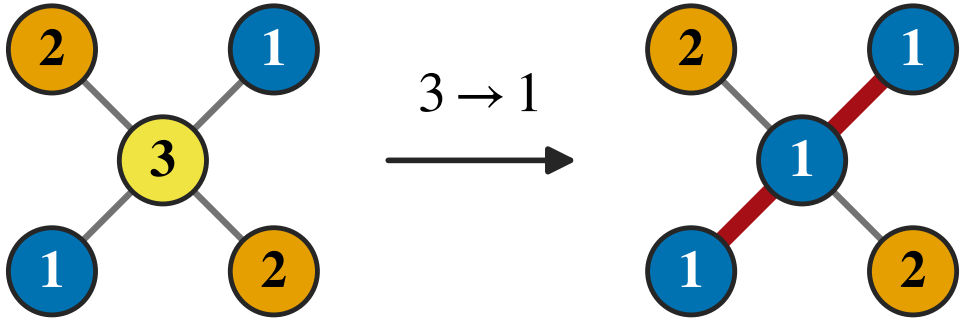


two states
$E = 1$

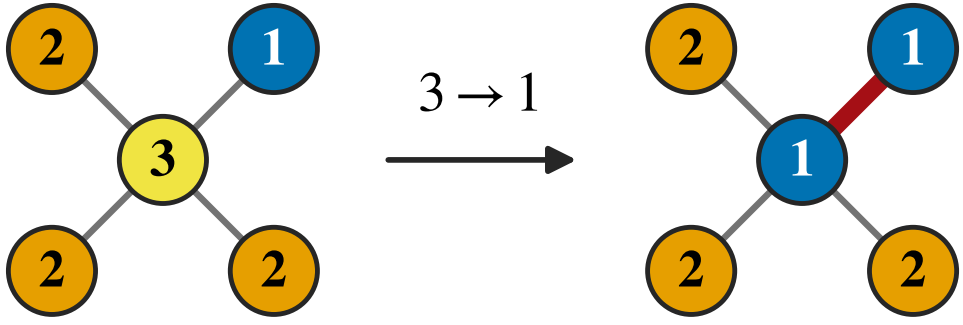


third state
$E = D_3$

(b)



$n_1 : n_2 = 2 : 2$
$\Delta E = 2 - D_3$



$n_1 : n_2 = 1 : 3$
$\Delta E = 1 - D_3$